NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Mathematics

MT-116

Calculus & Analytical Geometry

15-Week Course Plan

Course Level:	First Year Undergraduate
Duration:	15 Weeks
Course Type:	Mathematics Foundation Course
Target Students:	BSCS / Computer Engineering / Engineering Programs

Academic Year: 2026

1. Course Description

MT-116, *Calculus & Analytical Geometry*, provides students with the mathematical foundations required for further study in mathematics, computer science, computer engineering, and other engineering disciplines.

The course develops mathematical thinking progressively, beginning with numbers, sets, functions, and limits, and then introducing differential calculus, integral calculus, sequences and series, analytical geometry, and complex numbers.

Particular emphasis is placed on connecting mathematical concepts with applications in science, engineering, computing, optimization, numerical methods, and modern technology.

2. Course Philosophy

The course is organized around the following mathematical progression:

$$\boxed{\text{Numbers} \longrightarrow \text{Sets} \longrightarrow \text{Functions} \longrightarrow \text{Limits}}$$

$$\boxed{\text{Limits} \longrightarrow \text{Derivatives} \longrightarrow \text{Optimization} \longrightarrow \text{Approximation}}$$

$$\boxed{\text{Integration} \longrightarrow \text{Sequences} \longrightarrow \text{Infinite Series}}$$

$$\boxed{\text{Vectors} \longrightarrow \text{3D Geometry} \longrightarrow \text{Complex Numbers}}$$

The objective is not merely to cover individual chapters but to help students understand how different mathematical ideas are connected.

3. Course Learning Objectives

By the end of the course, students should be able to:

LO1: Explain and use fundamental properties of real numbers, sets, functions, and inequalities.

LO2: Analyze and sketch graphs of standard and composite functions.

LO3: Evaluate limits and determine continuity or discontinuity of functions.

LO4: Differentiate functions using standard differentiation rules and higher-order derivatives.

LO5: Apply differential calculus to optimization, curve analysis, asymptotes, and related problems.

LO6: Construct and use Taylor and Maclaurin approximations.

LO7: Solve nonlinear equations numerically using the Newton–Raphson method.

LO8: Calculate partial derivatives and analyze functions of several variables.

LO9: Solve unconstrained and constrained optimization problems using Lagrange multipliers.

LO10: Evaluate indefinite and definite integrals using appropriate integration techniques.

LO11: Apply integration to mathematical and engineering problems.

LO12: Analyze the convergence of sequences and infinite series using appropriate convergence tests.

LO13: Work with vectors, lines, planes, spheres, curves, and surfaces in three-dimensional space.

LO14: Represent and analyze complex numbers geometrically and algebraically.

LO15: Recognize the role of calculus and analytical geometry in modern scientific, engineering, and computational applications.

4. 15-Week Course Schedule

Table 1: Sequential 15-Week Course Plan for MT-116

Week	Main Topic	Key Concepts / Skills	Suggested Application
1	Numbers, Approximation & Inequalities	Rational, irrational and real numbers; rounding; significant figures; absolute value; quadratic and rational inequalities	Measurement, numerical error, engineering tolerances
2	Sets, Relations & Functions	Sets, operations, Venn diagrams, De Morgan's laws, Cartesian product, relations, functions and types	Data structures and mathematical models
3	Functions & Graphs	Absolute value, greatest integer, composite functions, inverse functions, transformations and standard graphs	Modeling real-world relationships
4	Limits & Continuity	Numerical, graphical and algebraic limits; one-sided limits; infinite limits; continuity and discontinuity	Motion, simulation and approximation
5	Differentiation	Definition of derivative; differentiation rules; product, quotient and chain rules; higher derivatives; Leibniz theorem	Velocity, acceleration and rates of change
6	Applications of Derivatives	Increasing/decreasing functions; extrema; first and second derivative tests; concavity; asymptotes; curve sketching; optimization	Engineering design and optimization
7	Taylor & Maclaurin Series	Taylor theorem; Maclaurin theorem; remainder forms; power series; standard expansions	Numerical approximation and computation
8	Midterm & Numerical Methods	Course review; Newton–Raphson method; nonlinear equations; error and convergence	Numerical solution of nonlinear equations
9	Functions of Several Variables	Functions of two and three variables; partial derivatives; higher-order partial derivatives; multivariable chain rule	Temperature, pressure and cost models

Week	Main Topic	Key Concepts / Skills	Suggested Application
10	Multivariable Optimization	Critical points; Hessian/second derivative test; maximum and minimum values; constrained optimization; Lagrange multipliers	Resource allocation and engineering design
11	Integral Calculus	Antiderivatives; substitution; integration by parts; partial fractions; reduction formulae	Area, accumulation and physical applications
12	Definite Integrals & Special Functions	Definite integrals; Fundamental Theorem of Calculus; convergence; applications; Beta and Gamma functions	Probability and engineering models
13	Sequences & Infinite Series	Sequences; convergence; comparison, ratio, root, Raabe and Gauss tests	Approximation and numerical computation
14	Analytical Geometry	Vectors; scalar and vector products; 3D coordinates; lines, planes, spheres; curves, surfaces and coordinate transformations	Computer graphics, robotics and CAD
15	Complex Numbers	Argand diagram; De Moivre formula; roots of polynomial equations; complex-plane regions; exponential, circular and hyperbolic functions	Signal processing, electrical engineering and control

5. Detailed Weekly Plan

5.1. Week 1 – Numbers, Approximation and Inequalities

Topics

- Rational, irrational and real numbers.
- Real number line and intervals.
- Rounding numerical values.
- Decimal places and significant figures.
- Absolute and relative error.
- Quadratic inequalities.
- Rational inequalities.
- Modulus inequalities.
- Graphical representation of inequalities.

Suggested Activity

Give students several numerical measurements and ask them to report the values to different numbers of significant figures. Discuss how rounding affects engineering calculations.

Key Mathematical Idea

$$\text{Exact Value} \longrightarrow \text{Approximation} \longrightarrow \text{Error Analysis}$$

5.2. Week 2 – Sets, Relations and Functions

Topics

- Definition of sets.
- Subsets and universal sets.
- Union, intersection and complement.
- Venn diagrams.
- De Morgan's laws.
- Cartesian product.
- Relations.
- Definition of a function.
- Domain and range.
- Types of functions.

Suggested Activity

Students identify examples of sets and relations from everyday life, such as students enrolled in different courses or relationships between input and output variables.

Key Mathematical Idea

$$\boxed{\text{A function describes a mathematical input-output relationship.}}$$

5.3. Week 3 – Functions and Graphs

Topics

- Absolute value function.
- Greatest integer function.
- Composite functions.
- Inverse functions.
- Transformations of functions.
- Standard algebraic functions.
- Exponential and logarithmic functions.
- Trigonometric functions.

- Graphing functions.

Important examples include

$$|x|, \qquad \lfloor x \rfloor, \qquad x^2, \qquad \sqrt{x}, \qquad e^x, \qquad \ln x, \qquad \sin x, \qquad \cos x.$$

Suggested Activity

Ask students to construct the graph of

$$f(x) = |x-2| + 1$$

starting from the basic graph

$$y = |x|.$$

5.4. Week 4 – Limits and Continuity

Topics

- Concept of a limit.
- Numerical approach to a limit.
- Graphical interpretation.
- Algebraic evaluation of limits.
- One-sided limits.
- Infinite limits.
- Limits at infinity.
- Continuity.
- Discontinuity.
- Types of discontinuities.

Introduce the fundamental notation

$$\lim_{x \to a} f(x).$$

Suggested Discussion

Ask students:

> What happens to a function when x gets closer and closer to a particular value?

This question naturally motivates the concept of a limit.

5.5. Week 5 – Differential Calculus

Topics

- Definition of derivative.
- Derivative as rate of change.
- Differentiation rules.
- Product rule.
- Quotient rule.
- Chain rule.
- Higher-order derivatives.
- Successive differentiation.
- Leibniz theorem.

Begin with

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Then develop standard differentiation rules.

Applications

$$\text{Position} \longrightarrow \text{Velocity} \longrightarrow \text{Acceleration}.$$

5.6. Week 6 – Applications of Derivatives

Topics

- Increasing and decreasing functions.
- Critical points.
- Local extrema.
- Global extrema.
- First derivative test.
- Second derivative test.
- Concavity.
- Points of inflection.
- Asymptotes.
- Curve sketching.
- Optimization.

Suggested Application

Optimization of a container or engineering component subject to material constraints.

The mathematical structure is

$$\text{Physical Problem} \longrightarrow \text{Mathematical Model} \longrightarrow \text{Objective Function} \longrightarrow \text{Optimization.}$$

5.7. Week 7 – Taylor and Maclaurin Theorems

Topics

- Taylor theorem.
- Maclaurin theorem.
- Lagrange form of the remainder.
- Cauchy form of the remainder.
- Power series.
- Taylor series.
- Maclaurin series.
- Error and approximation.

Important examples:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots .$$

Modern Application

Explain that computers often approximate complicated functions using polynomial or series expansions.

5.8. Week 8 – Midterm and Newton–Raphson Method

First Part

- Review of Weeks 1–7.
- Midterm examination.

Second Part

Introduce numerical solution of nonlinear equations.

For

$$f(x) = 0,$$

the Newton–Raphson iteration is

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}.$$

Example

To solve

$$x^2 - 2 = 0,$$

choose

$$x_0 = 1.$$

Then

$$x_1 = 1.5,$$

$$x_2 = 1.416666\ldots,$$

$$x_3 = 1.414215\ldots$$

and therefore

$$\sqrt{2} \approx 1.41421356.$$

Suggested Activity

Students implement the Newton–Raphson method using a calculator, spreadsheet, Python, MATLAB, or another computational tool.

5.9. Week 9 – Functions of Several Variables

Make the conceptual transition

$$y = f(x)$$

to

$$z = f(x, y).$$

Topics

- Functions of two and three variables.
- Domain and range.
- Graphs of multivariable functions.
- Partial derivatives.
- Higher-order partial derivatives.
- Mixed partial derivatives.
- Chain rule for several variables.

Examples:

$$T = T(x, y, z)$$

for temperature, or

$$C = C(x, y)$$

for a cost function.

Introduce

$$\frac{\partial f}{\partial x}, \qquad \frac{\partial f}{\partial y}.$$

5.10. Week 10 – Multivariable Optimization

Topics

- Critical points of functions of two variables.
- Second derivative test.
- Maximum and minimum values.
- Unconstrained optimization.
- Constrained optimization.
- Lagrange multipliers.

For unconstrained optimization,

$$\nabla f = 0.$$

For constrained optimization,

$$g(x, y) = c,$$

and

$$\boxed{\nabla f = \lambda \nabla g.}$$

Suggested Applications

- Resource allocation.
- Engineering design.
- Minimum cost.
- Maximum performance.
- Optimization in machine learning.

5.11. Week 11 – Integral Calculus

Topics

- Antiderivatives.
- Indefinite integrals.
- Basic integration rules.
- Substitution.
- Integration by parts.
- Partial fractions.
- Reduction formulae.

Introduce the conceptual relationship

$$\boxed{\text{Differentiation} \longleftrightarrow \text{Integration.}}$$

Emphasize the idea that integration represents accumulation.

5.12. Week 12 – Definite Integrals and Special Functions

Topics

- Definite integrals.
- Fundamental Theorem of Calculus.
- Convergence of definite integrals.
- Applications of definite integrals.
- Beta function.
- Gamma function.
- Important identities.

The Beta function is defined by

$$B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx,$$

while the Gamma function is

$$\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}\,dx.$$

A fundamental identity is

$$\Gamma(n+1) = n\Gamma(n).$$

For positive integers,

$$\Gamma(n) = (n-1)!.$$

5.13. Week 13 – Sequences and Infinite Series

Topics

- Sequences.
- Limits of sequences.
- Infinite series.
- Convergence and divergence.
- Comparison test.
- Ratio test.
- Root test.
- Raabe's test.
- Gauss test.

Introduce

$$a_1, a_2, a_3, \ldots$$

and

$$\sum_{n=1}^{\infty} a_n.$$

Emphasize that students should learn not only how to apply a test, but also how to decide **which convergence test is appropriate**.

5.14. Week 14 – Analytical Geometry

Topics

- Scalars and vectors.
- Vector addition.
- Dot product.
- Cross product.
- Three-dimensional coordinate system.
- Equation of a straight line.
- Equation of a plane.
- Sphere.
- Curves in three dimensions.
- Surfaces.
- Coordinate transformations.
- Curve and surface tracing.

For vectors

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k},$$

introduce the dot product

$$\mathbf{a} \cdot \mathbf{b}$$

and cross product

$$\mathbf{a} \times \mathbf{b}.$$

Modern Applications

- Computer graphics.
- 3D animation.
- Robotics.
- CAD.
- Computer vision.
- Game development.

5.15. Week 15 – Complex Numbers

Topics

- Definition of complex numbers.
- Algebra of complex numbers.
- Argand diagram.
- Modulus and argument.
- Polar representation.
- De Moivre's theorem.
- Roots of polynomial equations.
- Curves and regions in the complex plane.
- Exponential functions.
- Circular functions.
- Hyperbolic functions.
- Inverse functions.

For

$$z = a + ib,$$

the modulus is

$$|z| = \sqrt{a^2 + b^2}.$$

The polar representation is

$$z = r(\cos\theta + i\sin\theta).$$

De Moivre's theorem gives

$$\boxed{(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta).}$$

Modern Applications

- Electrical engineering.
- Signal processing.
- Communication systems.
- Control systems.
- Fourier analysis.

6. Mathematical Narrative of the Course

The complete course can be viewed as one mathematical journey.

$$\boxed{\text{Numbers}}$$

$$\Downarrow$$

$$\boxed{\text{Sets and Functions}}$$

$$\Downarrow$$

$$\boxed{\text{Limits}}$$

$$\Downarrow$$

$$\boxed{\text{Differentiation}}$$

$$\Downarrow$$

$$\boxed{\text{Optimization}}$$

$$\Downarrow$$

$$\boxed{\text{Taylor Approximation}}$$

$$\Downarrow$$

$\boxed{\text{Numerical Methods}}$

$\Downarrow$

$\boxed{\text{Multivariable Calculus}}$

$\Downarrow$

$\boxed{\text{Integration}}$

$\Downarrow$

$\boxed{\text{Infinite Series}}$

$\Downarrow$

$\boxed{\text{Vectors and 3D Geometry}}$

$\Downarrow$

$\boxed{\text{Complex Numbers}}$

7. Recommended Teaching Approach

For first-year students, each topic should ideally follow the sequence:

$$\boxed{\text{Concept} \rightarrow \text{Visualization} \rightarrow \text{Example} \rightarrow \text{Practice} \rightarrow \text{Application}}$$

Instead of presenting mathematics only as formulas, students should regularly be asked:

What does this mathematical object represent?

Why does this method work?

Where could we use this mathematics?

8. Suggested Modern Applications

Mathematical Topic	Modern Application
Functions	Computer modeling, programming, data analysis
Limits	Numerical approximation, continuous models
Derivatives	Optimization, motion, machine learning
Taylor Series	Numerical computation, approximation algorithms
Newton–Raphson	Root finding, engineering computation
Partial Derivatives	Machine learning, optimization, physics
Lagrange Multipliers	Resource allocation, constrained optimization
Integration	Area, probability, physics, engineering
Infinite Series	Approximation and numerical computation
Vectors	Computer graphics, robotics, computer vision
3D Geometry	CAD, gaming, animation, robotics
Complex Numbers	Signal processing, communications, electrical engineering

9. Suggested Assessment Structure

The following is a suggested structure and may be adjusted according to departmental policy.

Assessment Component	Suggested Weight
Quizzes	10%
Class Work / Tutorials	10%
Assignments	10%
Midterm Examination	25%
Final Examination	40%
Mathematical Activity / Project	5%
Total	**100%**

10. Suggested Weekly Structure

A typical teaching week can be organized as follows:

Session	Purpose
Lecture 1	Introduce the mathematical concept and motivation.
Lecture 2	Develop theory, formulas and worked examples.
Lecture 3	Applications, problem solving and conceptual discussion.
Tutorial / Class Work	Student-centered problem solving, activities and assessment.

11. Course Philosophy in One Diagram

$\boxed{\text{Describe the world}} \longrightarrow \boxed{\text{Understand change}} \longrightarrow \boxed{\text{Optimize systems}}$

$\boxed{\text{Approximate}} \longrightarrow \boxed{\text{Compute}} \longrightarrow \boxed{\text{Model}} \longrightarrow \boxed{\text{Solve}}$

Mathematics is not a collection of formulas.

It is a language for describing, analyzing, approximating, and solving problems.

12. End of Course

At the end of MT-116, students should be able to see the progression

$\boxed{\text{Elementary Mathematics} \rightarrow \text{Calculus} \rightarrow \text{Multivariable Mathematics} \rightarrow \text{Analytical Geometry} \rightarrow \text{Complex Mathe}}$

and recognize that these concepts provide the mathematical foundation for subsequent courses in mathematics, computer science, computer engineering, physics, statistics, and other engineering disciplines.

MT-116 – Calculus & Analytical Geometry

From mathematical concepts to mathematical thinking.